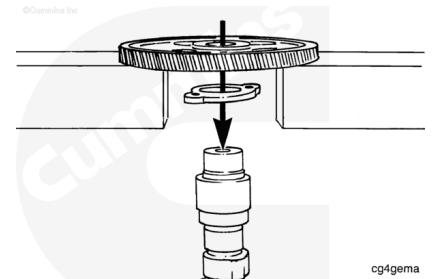


Trainings ▾

Disassemble

⚠ CAUTION ⚠

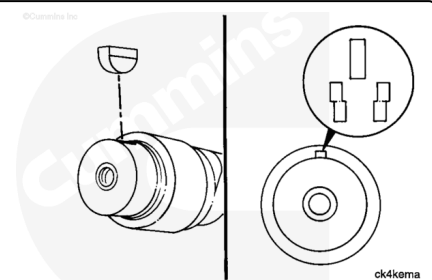
Do not use a heating torch to remove the camshaft gear. If a heating torch is used to remove the gear, the gear must be replaced. The heat of the torch will damage the heat treatment of the gear.



Use a hydraulic press to push the camshaft through the gear.

Camshaft keys are available in different sizes (amount of offset). The injection timing is controlled by:

- The selection of the camshaft key
- The direction of the key offset when compared to the camshaft gear rotation
- The amount of offset.



Check and record the direction of the key offset when compared to the camshaft gear rotation (same or opposite).

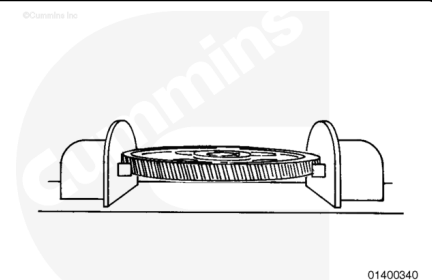
Remove the key.

Note : Engines with electronically actuated injectors do not use offset keys.

Magnetic Crack Inspect

⚠ CAUTION ⚠

Use a copper braid contact that has neoprene covers to reduce the possibility of burning the teeth of the gear.



If the gear contains a keyway, position the gear so that the keyway is pointing toward one of the contacts.

Use a magnetic particle testing machine.

Use the residual method and apply head shot amperage.

Adjust the amperage to the value specified in the table.

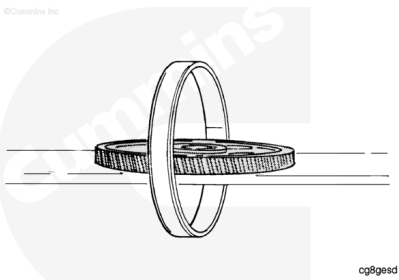
Amperage for Gear Sizes	
Gear Outside Diameter	Direct Current Amperage
Less than 101 mm [4 in]	1000
101 to 203 mm [4 to 8 in]	1500
Greater than 203 mm [8 in]	2000

Check the camshaft gear for cracks.

If the gear contains a keyway, position the gear so that the keyway is pointing toward the coil.

Use the residual method and apply coil shot amperage.

Adjust the amperage to the value specified in the table.



Amperage for Gear Sizes	
Gear Outside Diameter	Ampere Turns
Less than 101 mm [4 in]	4000
101 to 203 mm [4 to 8 in]	6000
Greater than 203 mm [8 in]	8000

Ampere turn is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

Check the camshaft gear fro cracks.

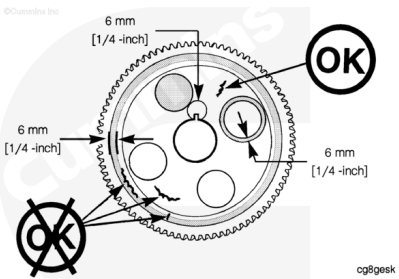
⚠ CAUTION ⚠

The gear must be demagnetized completely and cleaned thoroughly. Any small particles will cause engine damage.

An open indication is visible after the wetting operation has been completed. An indication below the surface is **not** visible after the wetting operation has been completed. An indication below the surface is visible with the use of the ultraviolet light.

Do **not** use the gear if:

- There is an open indication.
- There is an indication below the surface that is in the shaded area, shown in the graphic.
- There is an indication below the surface that is longer than 6 mm [0.250 in].



- There is a open indication that is in a circumferential direction.
- There is an open indication that is longer than 9.5 mm [0.375 in].
- There is an indication below the surface that is in the shaded area shown in the graphic.

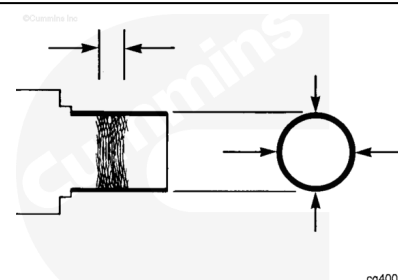
An indication below the surface in a radial direction is acceptable if it is in the area that is **not** shaded.

Demagnetize the gear.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the camshaft and gear with solvent and dry with compressed air.

Check the camshaft for fretting damage. The camshaft **must** be replaced if the fretting damage is more than 3 mm [0.125 in] wide.

Measure the outside diameter of the camshaft.

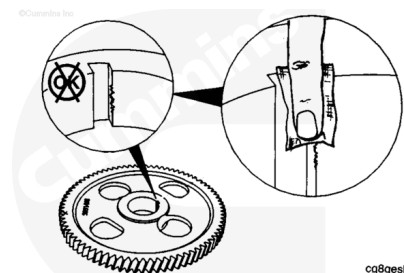
mm		in
86.90	MIN	3.421
86.99	MAX	3.425

If the camshaft is **not** within specifications. The camshaft **must** be replaced.

Check the keyway of the camshaft gear for burrs.

Remove any burrs with a fine emery cloth.

If the burrs can **not** be removed with a fine emery cloth or the keyway is damaged the camshaft gear **must** be replaced.



Inspect the bore of the gear for frets or burrs.

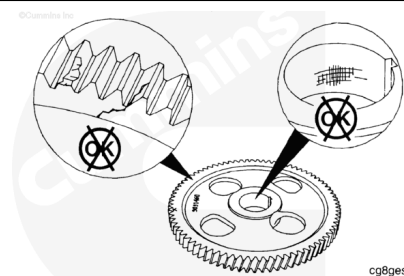
If fretting is in excess of 3 mm [0.125 in] wide the camshaft gear **must** be replaced.

If the fretting or burrs can **not** be removed with emery cloth, the camshaft gear **must** be replaced.

Measure the inside diameter of the camshaft gear.

mm		in
86.88	MIN	3.420
86.90	MAX	3.421

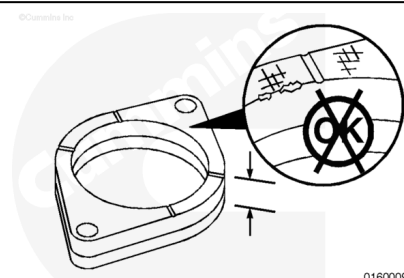
If the camshaft gear is **not** within specifications. The camshaft gear **must** be replaced.



Measure the camshaft thrust bearing thickness.

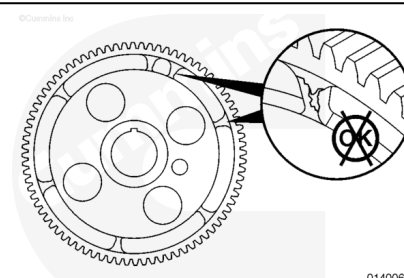
mm		in
13.95	MIN	0.549
14.05	MAX	0.553

If the camshaft thrust bearing is **not** within specifications, the camshaft thrust bearing **must** be replaced.



with Electronically Actuated Injector

Inspect the backside of the gear for cracks, chipped, or broken bosses that the cam position sensor reads. The gear **must** be replaced if the bosses are damaged.

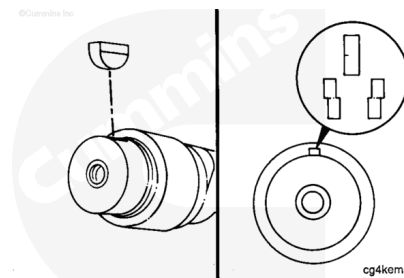


Assemble

Note : Engines with electronically actuated injectors do **not** use offset keys.

The camshaft gear **must** be removed to adjust the injection timing. If the camshaft is replaced, do **not** install the camshaft gear until the injection timing has been adjusted using a slip fit gear.

The QSK Timing Key Offsets table identifies the recommended camshaft key for each timing code. Machining tolerances on the camshaft and the gear can cause the injection timing **not** to be within specifications.



QSK Timing Key Offsets		
Key Part Number	Key Offset	Change in Push Rod Travel
216294	0.089 mm [0.004 in]	0.040 mm [0.002 in]
200711	0.178 mm [0.007 in]	0.079 mm [0.003 in]
3626781	0.229 mm [0.009 in]	0.102 mm [0.004 in]
216782	0.279 mm [0.011 in]	0.125 mm [0.005 in]
3626782	0.305 mm [0.012 in]	0.136 mm [0.006 in]
200709	0.381 mm [0.015 in]	0.170 mm [0.007 in]
3626783	0.432 mm [0.017 in]	0.192 mm [0.008 in]
200704	0.500 mm [0.020 in]	0.223 mm [0.009 in]
3626784	0.546 mm [0.021 in]	0.243 mm [0.010 in]
200708	0.584 mm [0.023 in]	0.260 mm [0.011 in]
3000491	0.686 mm [0.027 in]	0.306 mm [0.012 in]

QSK Timing Key Offsets

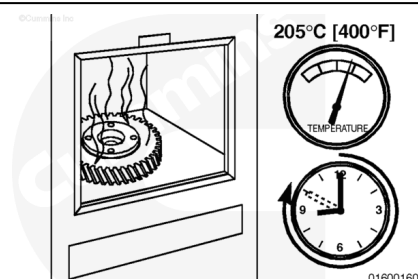
Key Part Number	Key Offset	Change in Push Rod Travel
200706	0.833 mm [0.033 in]	0.371 mm [0.015 in]
3000492	0.914 mm [0.036 in]	0.408 mm [0.016 in]
200714	0.991 mm [0.039 in]	0.441 mm [0.017 in]
3000493	1.092 mm [0.043 in]	0.487 mm [0.019 in]
3000494	1.194 mm [0.047 in]	0.532 mm [0.021 in]
3000495	1.295 mm [0.051 in]	0.577 mm [0.023 in]

⚠ CAUTION ⚠

Do not exceed the specified time or temperature. Damage to the gear and the gear teeth will result.

⚠ CAUTION ⚠

To reduce the possibility of gear damage, do not attempt to install the gear without using heat.



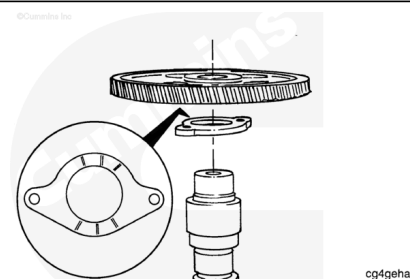
Use an oven with the temperature adjusted to 205°C [400°F]. Heat the gear in the oven for a **minimum** of 40 minutes but **not** more than a **maximum** of one hour. The inner diameter of the gear will become larger and will simplify the installation of the gear on the camshaft.

⚠ WARNING ⚠

To reduce the possibility of severe burns, wear protective gloves when installing the heated camshaft gear.

⚠ CAUTION ⚠

Allow the air to cool the gear. Do not use water or oil to reduce the cooling time. It can cause the gear to crack.



The timing marks on the camshaft gear **must** be visible from the front of the camshaft after the gear is installed. The keyway **must** be aligned with the key in the gear.

Install the thrust bearing.

Remove the gear from the oven.

Install the camshaft gear onto the camshaft.